

SAHIL YADAV

AI Engineer · Agentic Systems & GenAI Backend Development

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PROFESSIONAL OBJECTIVE

Full-Stack Engineer and B.Tech AIML student with hands-on production experience building agentic AI systems. Designed and shipped Synapse, an asynchronous Python/FastAPI backend orchestrating LangChain and LangGraph agents with the Gemini Vision API and a local Retrieval-Augmented Generation (RAG) pipeline to convert unstructured documents into strongly-typed, Pydantic-validated JSON. Also solo-architected CampusHub, a live multi-module React/Node platform. Seeking an AI Engineer role to design, orchestrate, and scale production agentic AI systems.

TECHNICAL SKILLS

AI/ML, GenAI & Agentic Systems: Python, LangChain, LangGraph (multi-agent orchestration), Retrieval-Augmented Generation (RAG), Generative AI, Google Gemini Vision API, Scikit-learn, TensorFlow, Keras

Backend & APIs: FastAPI, Node.js, Express.js, RESTful API Design, Async Architecture, Pydantic v2 schema validation

Vector DB, Databases & Cloud: Vector Databases (ChromaDB/Pgvector), PostgreSQL, Neon Serverless DB, Supabase, Clerk Auth, Cloudinary CDN, MongoDB

DevOps & Tools: Docker, Git/GitHub, Vercel, Render, Postman, Linux CLI, VS Code

Frontend: React 18, Vite, JavaScript (ES6+), TypeScript, Tailwind CSS

Cybersecurity (applied research): Wireshark, Zeek, PCAP/malware analysis

PROFESSIONAL EXPERIENCE

Cybersecurity Research & Forensic Intern June 2026 – Present

IFSO (Intelligence Fusion & Strategic Operations) — Delhi Police, New Delhi

- Interning on real threat cases performing Android malware reverse engineering — wrote custom Python scripts for XOR decryption of obfuscated payloads and extracted live C2 API keys — and network forensics (Wireshark, Zeek, NetworkMiner) to trace infection vectors and attribute compromised hosts.

ENGINEERING PROJECTS

Synapse — AI-Driven Invoice, Knowledge & Ledger Extractor

Python · FastAPI · LangChain · LangGraph · Vector DB · Google Gemini Vision API · Pydantic v2 · Async Architecture

- Built an asynchronous Python/FastAPI backend running agentic AI workflows that process handwritten receipts and unformatted merchant invoices into structured financial data via a REST interface.
- Orchestrated multi-agent pipelines using LangChain and LangGraph to dynamically route complex, multi-page document payloads through extraction and verification stages without blocking the primary event loop.
- Developed a local RAG system backed by a Vector DB to index and query parsed document embeddings for instant contextual retrieval over previously processed documents.
- Enforced strict schema validation on all generative outputs using Pydantic v2 blueprints, transforming loose LLM text into strongly-typed JSON (vendor details, line items, totals, tax fields), eliminating downstream parsing errors.

CampusHub — Student Lifestyle Platform

React 18 · Node.js/Express · Neon PostgreSQL · Clerk Auth · Docker

- Architected and solo-deployed a live multi-module platform (P2P trading, notes, events, hackathon teams); designed RESTful Node/Express APIs and containerized the stack with Docker for local/prod parity.

Data Science & Deep Learning Pipelines

Python · TensorFlow · Keras · Scikit-learn

- Built a Scikit-learn/NLTK sentiment classifier (TF-IDF vectorization) and a CNN image classifier on CIFAR-10 using TensorFlow/Keras with data augmentation and MobileNetV2 transfer learning.

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning 2024 – 2028 (Expected)

Vivekananda Institute of Professional Studies (VIPS), New Delhi · 2nd Year

CERTIFICATIONS & TRAINING

- Fundamentals of Agentic AI — Microsoft Learn · 2025
- Production-Ready AI Agents Core Blueprint — GeeksforGeeks Engineering Hub · 2025